

Active Directory Home Lab - Phase 2

Installing Windows Server 2022 and Preparing DC1

Objective

Install Windows Server 2022, configure the DC1 server, and prepare the environment for Active Directory Domain Services.

1. Mount the Installation ISO

With the virtual machine powered off:

1. Select **DC1** in VirtualBox.
2. Click **Settings**.
3. Open **Storage**.
4. Select the empty optical drive.
5. Click the CD icon.
6. Choose **Choose a Disk File**.
7. Browse to and select the Windows Server 2022 ISO.

2. Start the Virtual Machine

When the machine boots:

1. Choose your preferred:
 - a. Language
 - b. Time and currency format
 - c. Keyboard layout
2. Click:

Next

3. Click:

Install Now

3. Select the Operating System

Choose:

Windows Server 2022 Standard Evaluation (Desktop Experience)

Important

Although Server Core is commonly used in production environments, the Desktop Experience edition provides a graphical interface and is easier to learn with.

4. Select Installation Type

Choose:

Custom: Install Windows only (Advanced)

5. Select the Installation Disk

You should see:

- Drive 0 Unallocated Space
- 80 GB

Highlight the drive and click:

Next

Windows will automatically create the required partitions.

6. Configure the Administrator Account

After installation completes:

1. Create a strong Administrator password.
2. Record the password securely in your notes.
3. Click **Finish**.

7. Log In

VirtualBox uses:

Right Ctrl + Delete

instead of:

Ctrl + Alt + Delete

Log in as:

Administrator

8. Rename the Server

Once Server Manager opens:

Navigate to:

Server Manager → Local Server → Computer Name → Change

Rename the server:

DC1

Click:

OK

Restart the server when prompted.

9. Configure a Static IP Address

Open:

Control Panel → Network and Internet → Network Connections

1. Right-click **Ethernet**.
2. Select **Properties**.
3. Double-click:

Internet Protocol Version 4 (TCP/IPv4)

IP Configuration

Setting	Value
IP Address	192.168.1.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
Preferred DNS Server	192.168.1.10

Note

The DNS server points to itself. This configuration will make more sense once Active Directory Domain Services are installed.

10. Verify Network Connectivity

Verify IP Configuration

Open Command Prompt and run:

```
ipconfig
```

Verify Internet Connectivity

Run:

```
ping google.com
```

Verify Local Connectivity

Run:

```
ping 192.168.1.10
```

11. Configure a Bridged Network Adapter

Before promoting DC1 to a Domain Controller, the VM should be connected to the physical network.

1. Power off DC1.
2. Select **DC1** in VirtualBox.
3. Click **Settings**.
4. Select **Network**.
5. Under Adapter 1, change:

NAT

to:

Bridged Adapter

This allows DC1 to function as another device on the home network and simplifies communication with future clients.

12. Saving the State of the Virtual Machine

When closing the virtual machine, VirtualBox provides several options:

- Save the machine state
- Send the shutdown signal
- Power off the machine

Recommended Option

Select:

Save the machine state

This preserves the current state of the virtual machine and allows it to resume exactly where it left off without performing a full shutdown and restart.

Note

Although saving the machine state is convenient during lab exercises, periodically performing a normal shutdown and reboot is recommended to ensure updates are applied properly and to verify that services start correctly after a restart.

Verification

Confirm the following:

- Windows Server 2022 installed successfully.
- Administrator account created.
- Server renamed to DC1.
- Static IP address assigned.
- Internet connectivity verified.
- Local connectivity verified.
- Network adapter changed to Bridged Adapter.

Lessons Learned

- Installing Windows Server 2022.
- Using Server Manager.
- Configuring a static IPv4 address.
- Understanding DNS settings.
- Testing network connectivity.
- Configuring VirtualBox networking modes.

Next Steps

Phase 3 - Promoting DC1 to a Domain Controller

- Install Active Directory Domain Services.
- Create the first domain.
- Configure DNS.
- Promote DC1 to a Domain Controller.